



Q1

Classify the main threats using the CIA triad and security goals

5 Marks

Objective: Identify which of Confidentiality, Integrity and Availability were violated in the breach and state the security goal affected by each attacker action.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	Read the breach summary and list every attacker action you can identify.	Breach report	List of actions
2	ANALYSE	For each action, decide whether it broke Confidentiality, Integrity or Availability.	CIA triad	C / I / A tag per action
3	ANALYSE	State the specific security goal that failed (e.g. "data secrecy", "data accuracy", "service uptime").	Security goals	Goal named per action
4	REPORT	Summarise which CIA element was most severely impacted overall.	—	One-line verdict
5	REPORT	Record the classification in a small table for the report.	Table template	Completed CIA table

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to Acme HealthCare

The attacker (1) stole 50,000 patient records, (2) encrypted the database with ransomware, and (3) took the booking portal offline. Each action is tagged against the CIA triad:

Attacker action	CIA element broken	Security goal failed
Stole patient records	Confidentiality	Data secrecy
Encrypted the database	Integrity	Data accuracy / access
Booking portal taken offline	Availability	Service uptime

Overall verdict: Confidentiality was most severely impacted, because sensitive medical data left the organisation permanently.

Reusable Template

Yahoo Data Breach:

First discovered in 2016, multiple breaches occurring between 2013 and 2014.

These attacks exposed the names, hashed password, phone numbers, email addresses, date of births and security questions of users.

Attacks have to occurred in august 2013 and late 2014, but yahoo disclosed the breaches only years later in 2016 and 2017. This Delay created further criticism surrounding the company's incident handling and transparency.

Data Breach Timeline:



- 2013 - First breach is believed to occur; 3 billion accounts compromised.
- 2014 – Second breach targets roughly 500 million accounts.
- September 2016 – Yahoo publicly discloses the 2014 attack.
- December 2017 – Yahoo acknowledges the 2013 breach, initially reporting 1 billion accounts impacted.
- October 2017 – Yahoo updates the 2013 breach impact to 3 billion accounts..

CIA classification table (copy and fill for your own breach)

Attacker action	CIA element (C / I / A)	Security goal failed
Spear-phished an employee to gain network access	Confidentiality	Credential secrecy
Stole the full user database (~3B accounts in 2013; ~500M in 2014 — names, emails, phone numbers, DOB, hashed passwords, security Q&A)	Confidentiality	Data secrecy
Forged authentication cookies via the compromised account management tool to access accounts without a password	Integrity	Authentication / session trust

Your Output / Screenshot (paste result here):

Overall verdict: Confidentiality was most severely impacted — this remains the largest data breach in history by accounts affected (all 3 billion Yahoo accounts). Integrity was also broken: forged cookies let attackers impersonate an estimated 32 million accounts without ever needing the real password, meaning Yahoo's authentication mechanism itself could no longer be trusted. Availability was not materially affected — Yahoo's services stayed online throughout.

- **Two separate incidents, one disclosure timeline:** The 2014 breach (500M accounts) was disclosed first, in September 2016; the earlier and larger 2013 breach (originally 1B+, later revised to all 3B accounts) was disclosed three months later, in December 2016, then corrected again in October 2017.
- **Attribution:** The US DOJ's March 2017 indictment named two alleged Russian FSB officers (Dmitry Dokuchaev, Igor Sushchin) and two hackers (Alexsey Belan, Karim Baratov) — tying the integrity-breaking cookie-forging technique specifically to Belan's access.
- **Why the cookie forgery mattered more than it sounds:** Because it bypassed passwords entirely, resetting passwords alone didn't fully remediate the breach — Yahoo also had to invalidate the forged cookies and the underlying account management tool logic.
- **Business impact:** Verizon cut its acquisition offer by \$350 million after the disclosures, before the deal closed in 2017.
- **Regulatory penalty:** The SEC fined Yahoo \$35 million in 2018 for delayed disclosure to investors.